

Robot Learning: From Fundamentals to Foundation Models

Lecture 11: Embodied Reasoning and Test-time Scaling

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04.05.2026

Is **Scaling Robot Data** All We Need?

What is Missing? **Intelligent Reasoning**



Robots capable of many tasks



Intelligent Reasoning: Generalization, Interpretability, Interaction, Controllability

What is Missing? **Intelligent Reasoning**

- Training LLMs on all of Internet not enough
- LLMs train specifically for **reasoning**



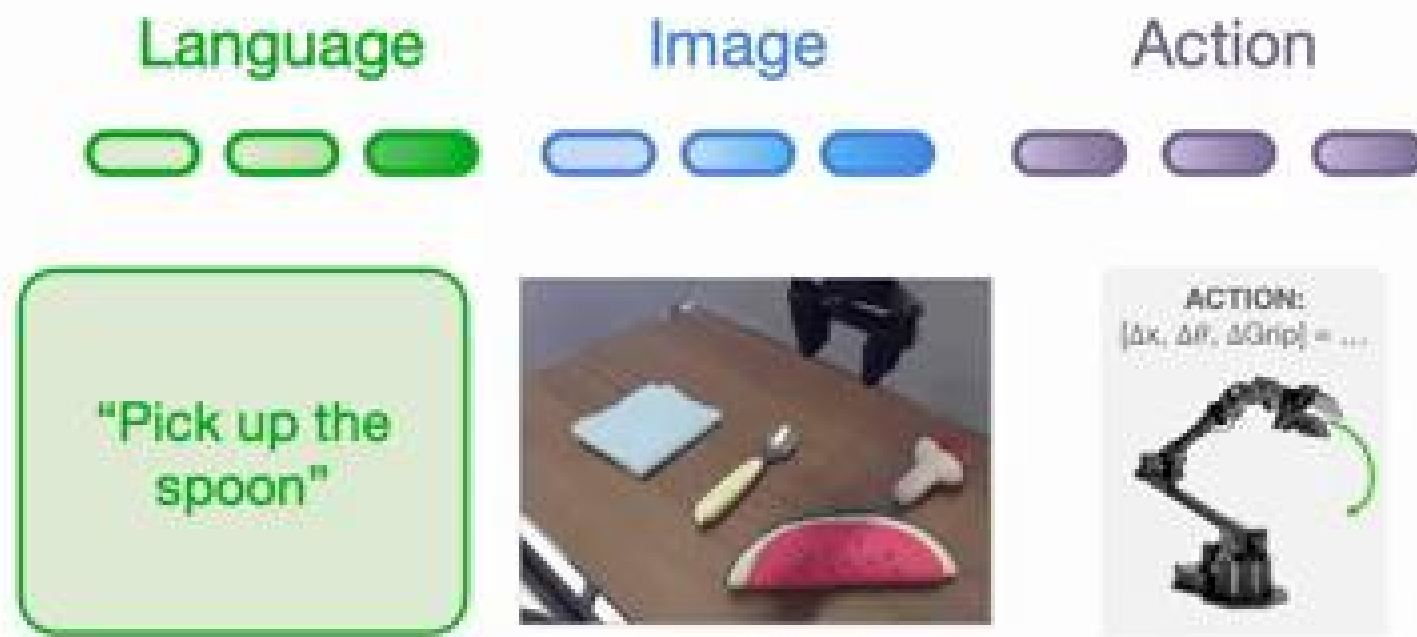
Robots capable of many tasks



Intelligent Reasoning: Generalization, Interpretability, Interaction, Controllability

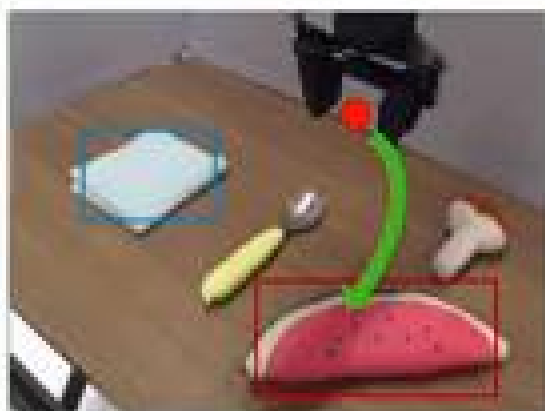
Robotics as Multimodal Sequence Modeling

- Can we **transfer** algorithmic improvements like **reasoning**?
- Can a robot **think harder** to **act better**?



Embodied Chain-of-Thought Reasoning

Break down hard problems into intermediate reasonings



P Put the watermelon on the table

R

Plan: 1. Move to watermelon, 2. Grasp it, ...
Subtasks: Move to watermelon
Move: Move backwards
Blocks: Gripper Pos

ACTION: [A₀, A₁, A₂] = ...

Standard policies directly map images to actions

✗ They thus often fail on novel tasks

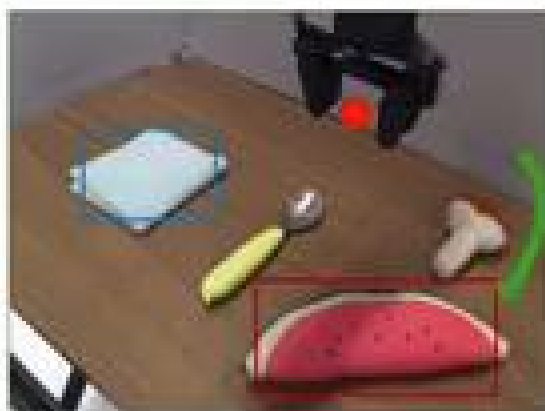
We thus propose to add embodied reasoning to the policy

Embodied reasoning causes policies to **"look"** and **"think"** carefully

✓ This yields better performance, interpretability, and generalization!

Embodied Chain-of-Thought Reasoning

Break down hard problems into intermediate reasonings



Put the watermelon on the towel

Plan: 1. Move to watermelon, 2. Grasp it, ...
Subtask: Move to watermelon
Move: Move backwards
Blocks: Gripper Pose

ACTION: [54, 18, 10] ...

Standard policies directly map images to actions

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We thus propose to add embodied reasoning to the policy

Creating **Multimodal Reasonings** at Scale

Maximally use existing data by **distilling foundation models**

How do we create robot reasoning data at scale?

We use a suite of foundation models to generate reasoning data for training



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Embodied Chain-of-Thought Dataset

Put the watermelon on the towel



Episode 5 (24, 40, 40ms)

REASONING

TASK:
Place the watermelon on the towel

PLAN:
1. Move to watermelon
2. Firmly grasp it
3. ...

SUBTASK: The watermelon is the first object the robot needs to interact with → Move to the watermelon

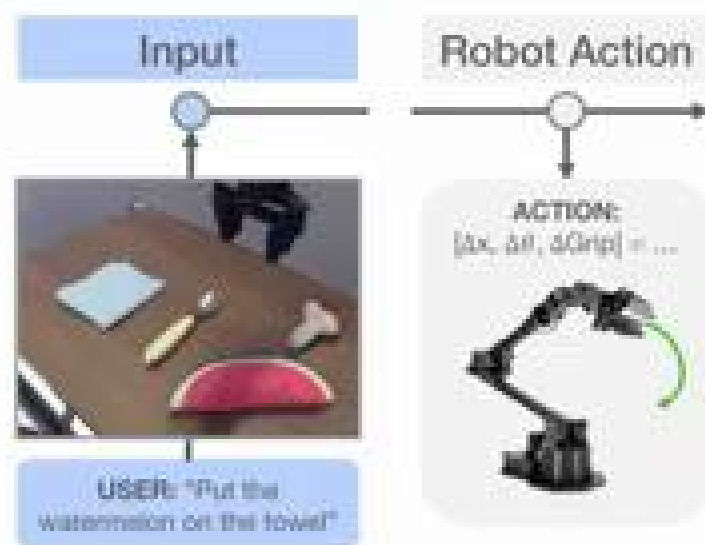
MOVE: The watermelon is behind the robot, so it needs to move backward → Move backward

GRIPPER POS:
[156, 50]

VISIBLE OBJs:
Watermelon [100, 140, 141, 125]
Towel [20, 50, 210, 100]
...

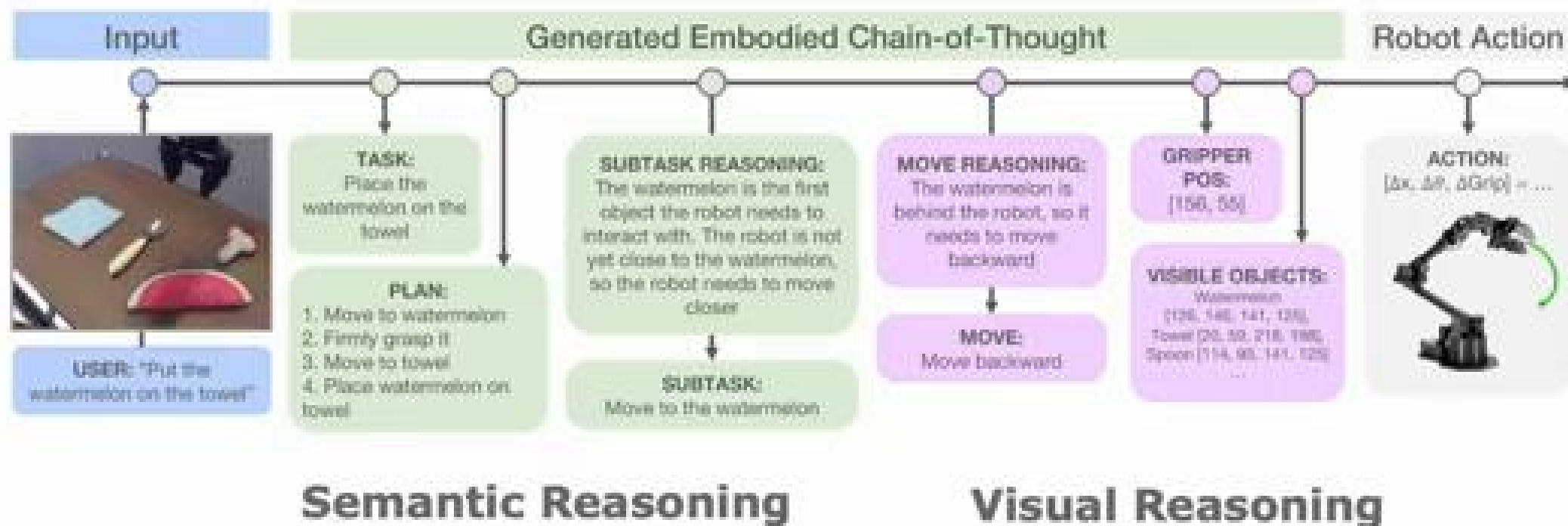
Embodied Chain-of-Thought Architecture

“Think carefully” and “look carefully” before acting



Embodied Chain-of-Thought Architecture

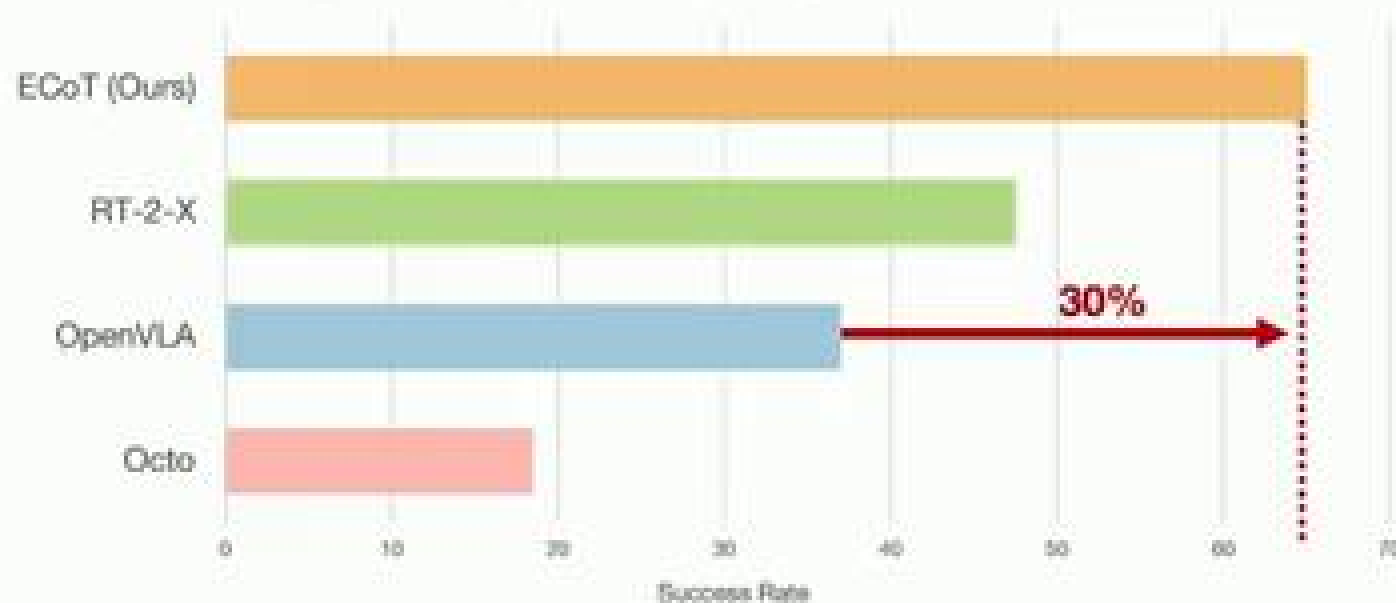
“Think carefully” and “look carefully” before acting



Generalization Evaluations

Boosts performance 30% **without extra robot data**

Our Embodied Chain-of-Thought VLA is more performant than other generalist policies, especially on novel tasks



Human-Robot **Interaction**

- ECoT makes policies more **interpretable**
- Allows human to **correct** behaviors



Lastly, human operators can correct robot behaviors by editing reasoning chains

Pick up any object that is not yellow

Plan: ...

Subtask: Move to the yellow cloth

Move: Move left ✓

Robot misbehaves due to incorrect reasoning

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Robot misbehaves due to incorrect reasoning

Pick up any object that is not yellow

Plan: ...

Subtask: Move to the yellow cloth

Move: Move left

ChatGPT's corrected reasoning
fixes robot behavior!

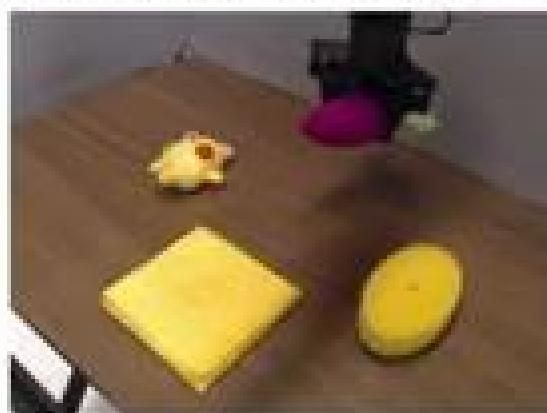
The eggplant near the arm is purple!

Correct

Human-Robot **Interaction**

- ECoT makes policies more **interpretable**
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Lastly, human operators can correct robot behaviors by editing reasoning chains



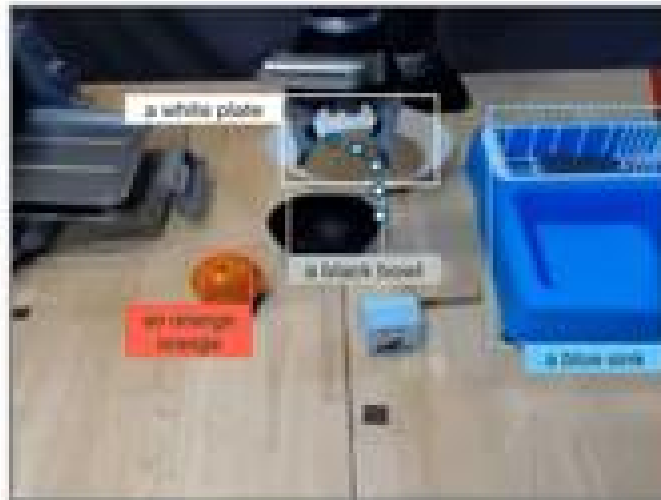
Pick up any object that is not yellow

Plan: ...

Subtask: The eggplant is a non-yellow object and should be picked up. → Move to the eggplant. ...

ChatGPT's corrected reasoning
fixes robot behavior!

Reasonings Generalize to Unseen Robots

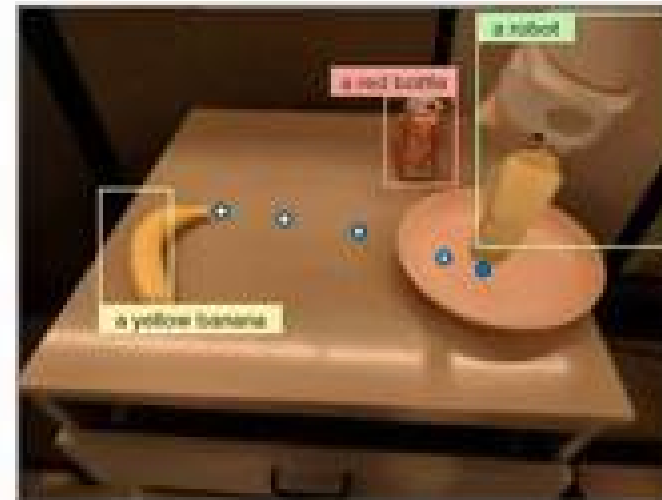


TASK: Put the black bowl in the blue sink.

PLAN: Move to the black bowl. Pick up the black bowl. Move to the sink. Drop the black bowl in the blue sink.

SUBTASK: The black bowl is the first object to interact with → Move to the black bowl.

MOVE: The black bowl is to the right of the robot → Move right.

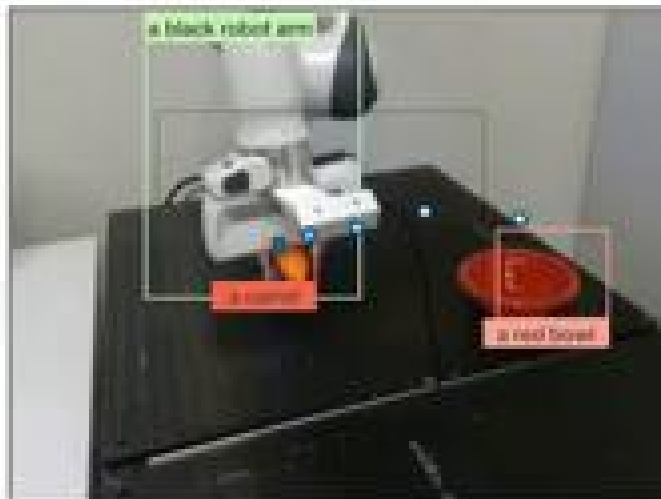


TASK: Move to the banana.

PLAN: Move to the banana, grasp the banana, move the banana to the plate, release the banana.

SUBTASK: The banana is on the left side of the plate, so the robot needs to move left → Move to banana.

MOVE: The robot needs to move left to get closer to the banana → Move left.



TASK: The task is to put the carrot in the red bowl.

PLAN: 1. Move to the carrot. 2. Grasp the carrot. 3. Move to the red bowl. 4. Release the carrot.

SUBTASK: The robot has grasped the carrot and needs to move it to the red bowl → Move to the red bowl.

MOVE: The red bowl is to the right of the robot, so the robot needs to move right to reach it → Move right.



TASK: Put the grey sponge on the yellow towel.

PLAN: Move to the sponge, pick up the sponge, move to the towel, put down the sponge.

SUBTASK: The sponge is on the table, and it needs to be reached first → Move to the sponge.

MOVE: The sponge is below the robot, so move down → Move down.

Impact of Embodied Chain-of-Thought

- Key: new standard to train VLAs with reasoning
- Improves generalization, interpretability & interactivensess
- Adoption by industry and academia



Gemini Robotics VLA 1.5

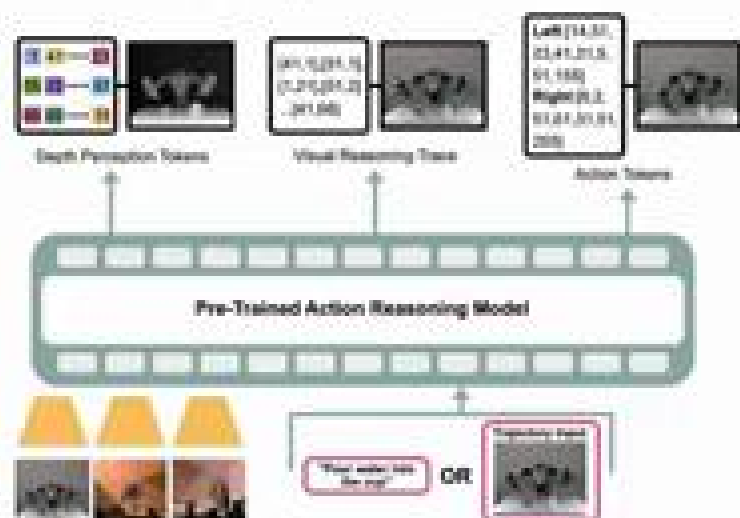
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Gemini Robotics VLA 1.5

Follow Up Works on Embodied Reasoning



MolmoAct: Action Reasoning Models that can Reason in Space
Lee et al., 2025.



Action-Free Reasoning for Policy Generalization
Clark et al., 2025.



CoT-VLA: Visual Chain-of-Thought Reasoning for Vision-Language-Action Models
Zhao et al., 2025.



Self-Supervised Bootstrapping of Action-Predictive Embodied Reasoning
Ganai et al., 2026.

Inference Cost of Reasoning

Autoregressive VLA Inference Speed
OpenVLA

4 Actions/Second

Embodied Chain-of-Thought Reasoning
Inference Speed

4 Seconds/Action



Can we get the benefits of reasoning
without the inference costs?

Why Does Embodied Reasoning Help Robot Policies?

1. Better representation learning

2. Improved learning "curriculum"

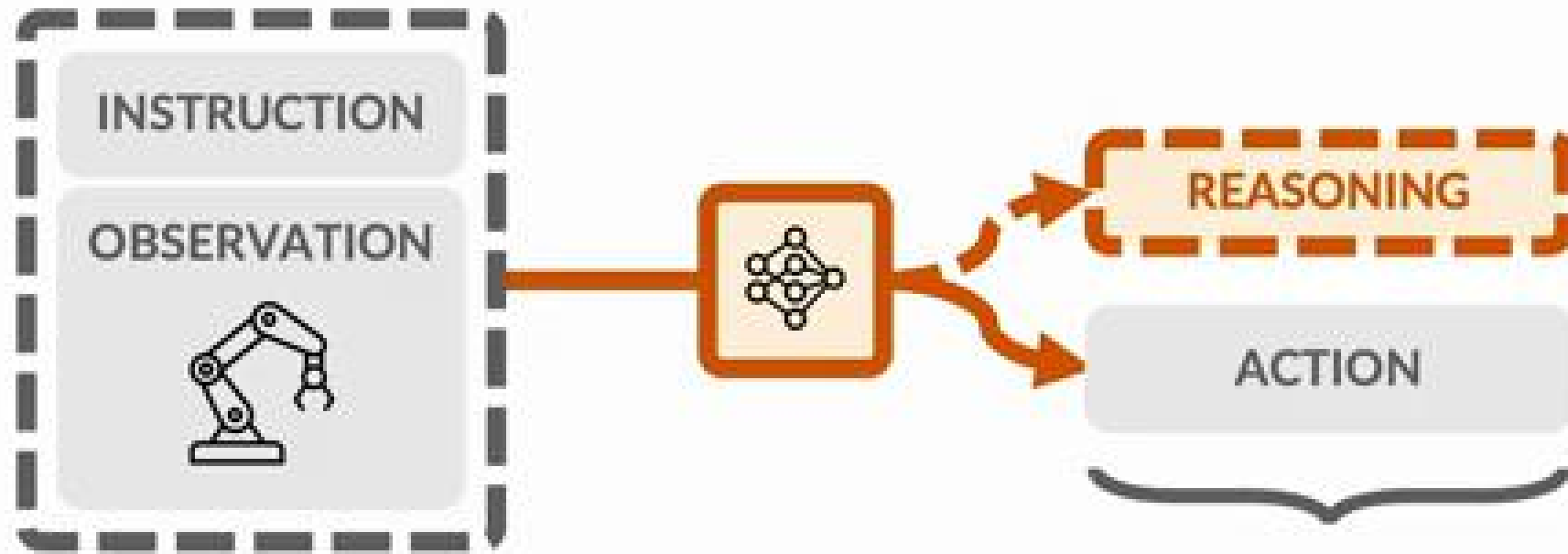
3. Increased policy expressivity

Why Does Embodied Reasoning Help Robot Policies?

2. Improved learning "curriculum"

3. Increased policy expressivity

1. Better representation learning



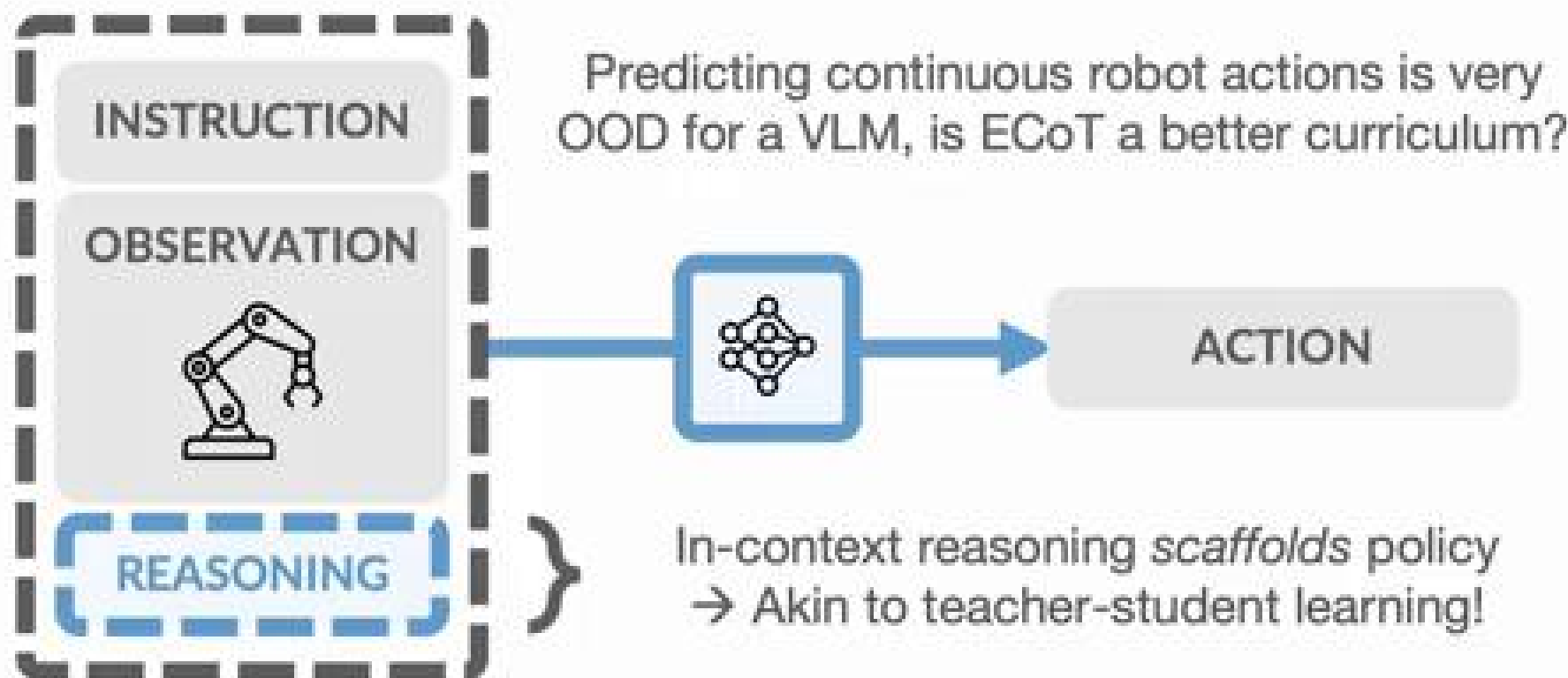
Train on both, but only choose
actions at test time

Why Does Embodied Reasoning Help Robot Policies?

1. Better representation learning

3. Increased policy expressivity

2. Improved learning "curriculum"

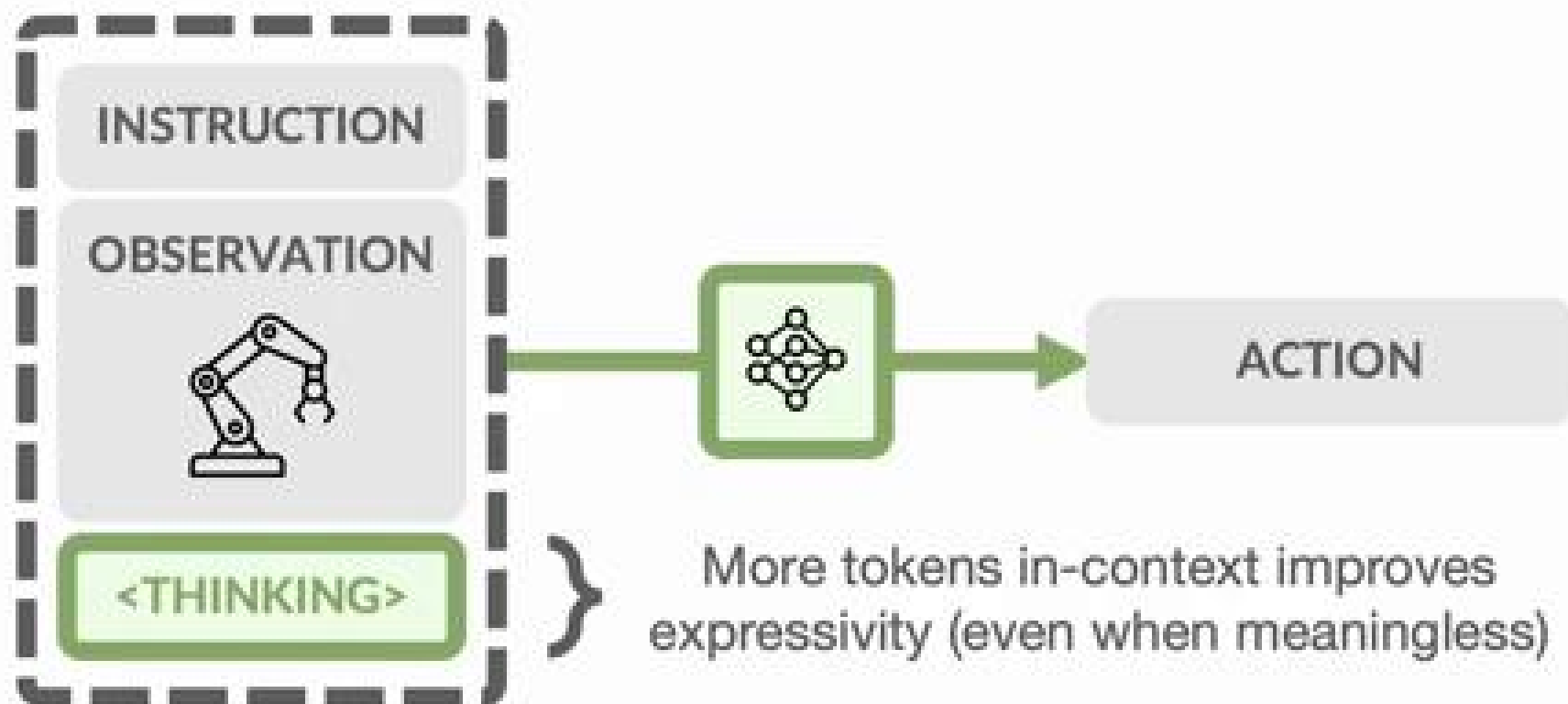


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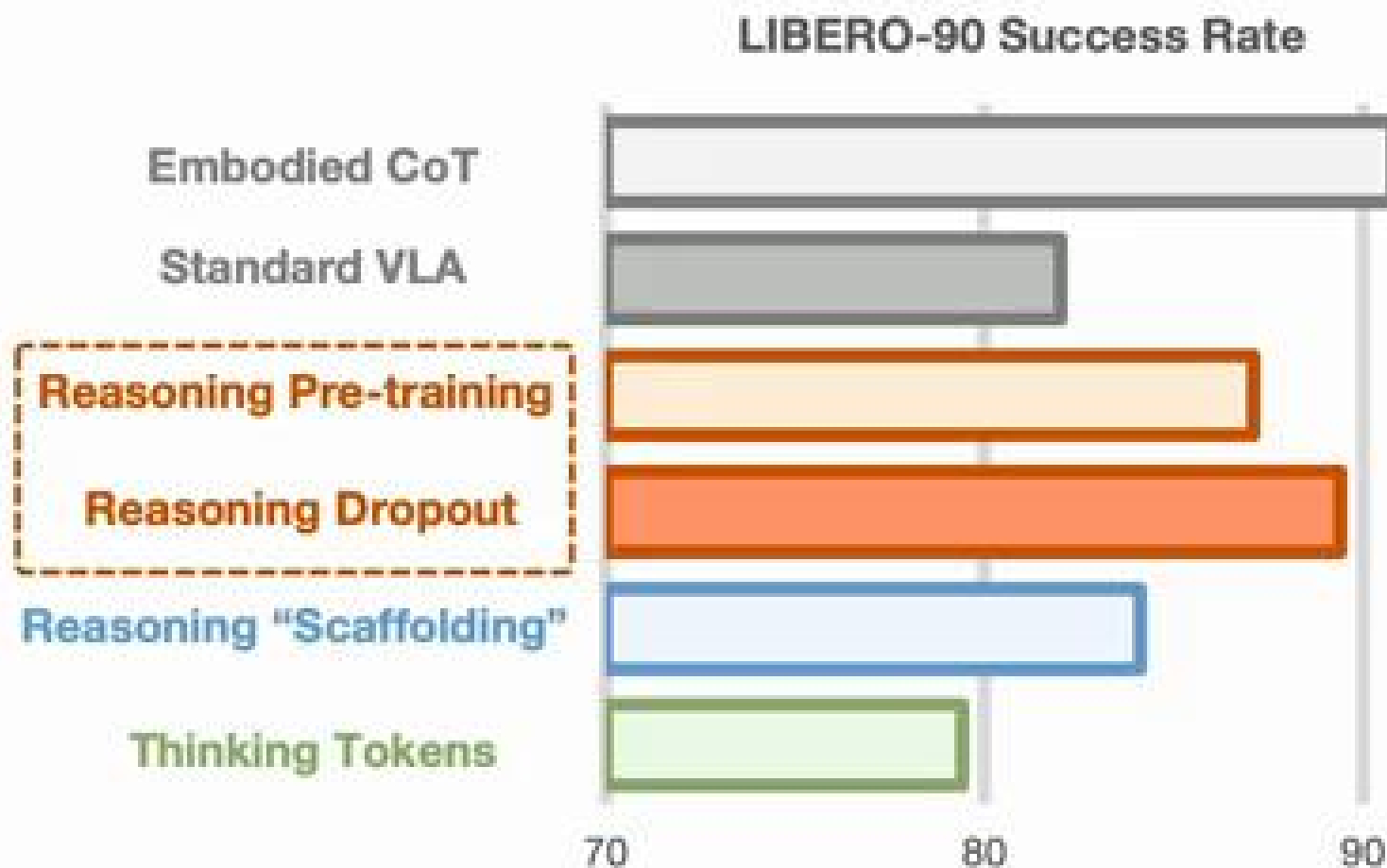
Let's Think Dot by Dot: Hidden Computation
in Transformer Language Models

Pfau et al, 2024.

Think before you speak: Training Language
Models With Pause Tokens

Goyal et al, 2024.

Results in LIBERO-90



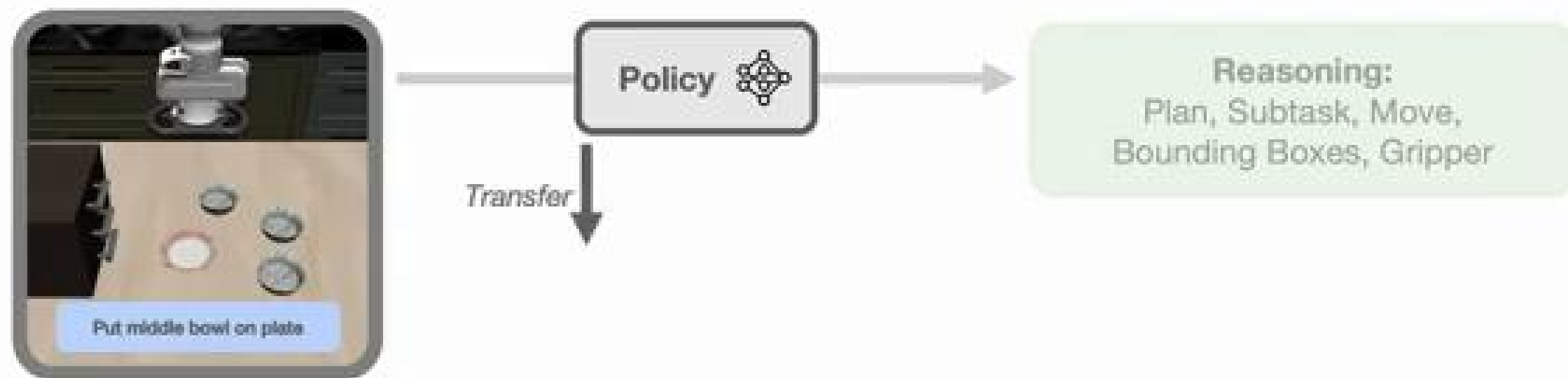
Training Strategies for Efficient Embodied Reasoning

Chen, Belkhale, Mirchandani, Mees, Driess, Pertsch, Levine. CoRL, 2025.

Embodied Chain-of-Thought Lite

Reasoning Dropout

Reasoning Pre-training



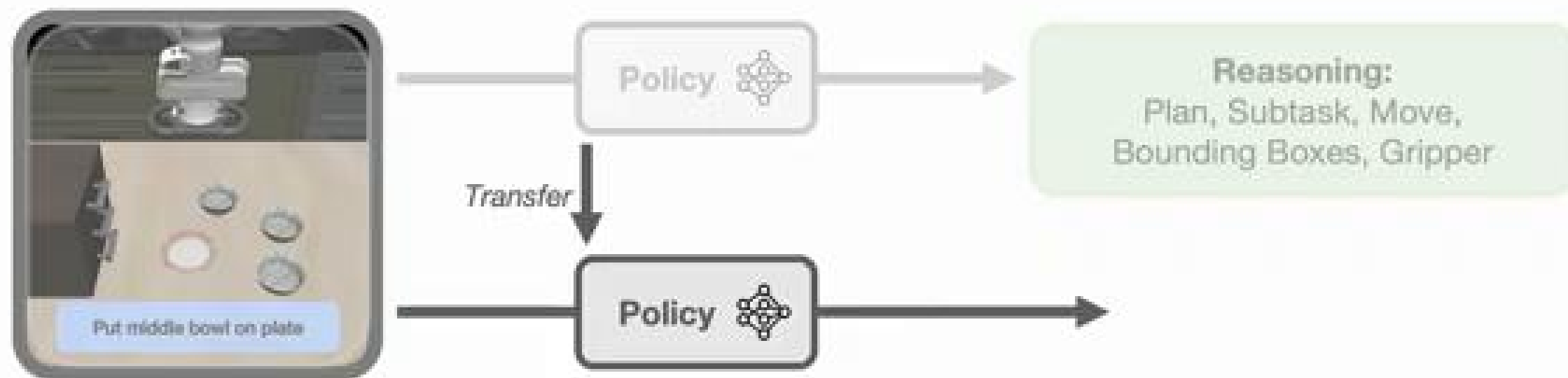
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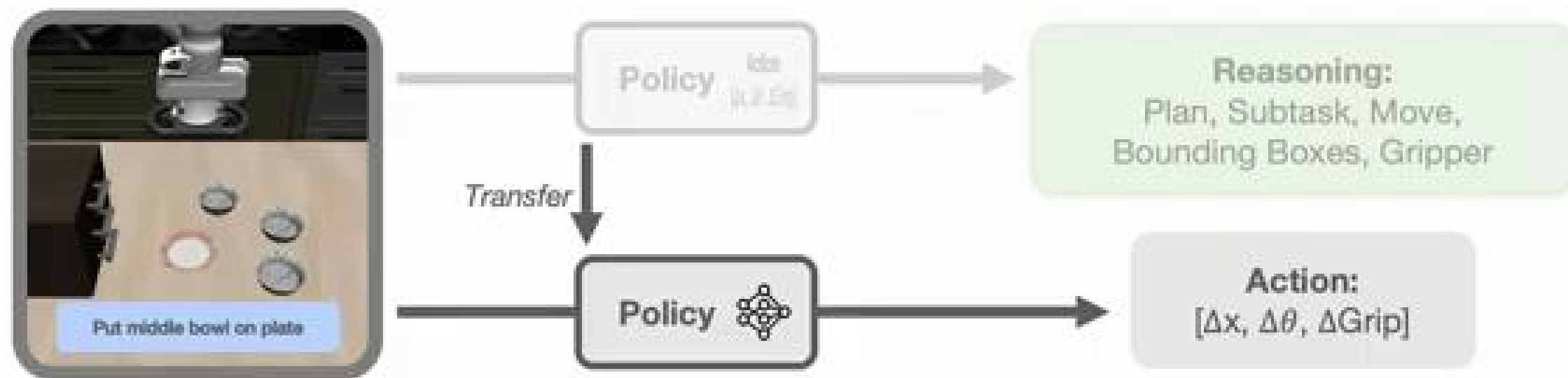
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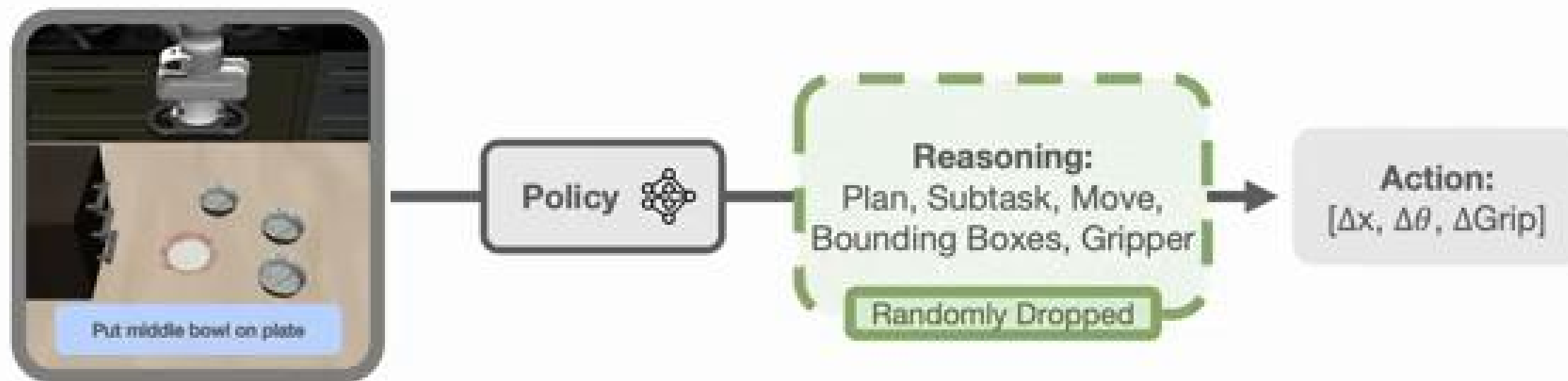
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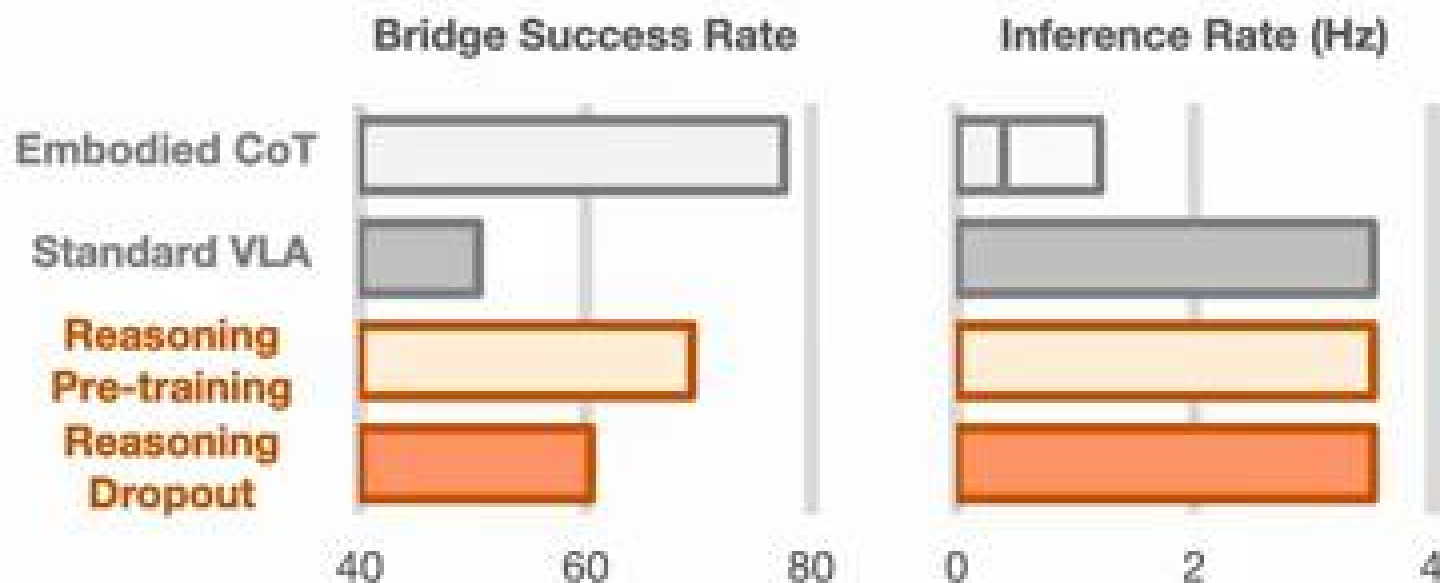
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Chen, Belkhale, Mirchandani, Mees, Driess, Pertsch, Levine. CoRL, 2025.

Results in Real-World Bridge WidowX Setup



ECoT-lite is faster than reasoning policies and stronger than non-reasoning policies



More Performant than Standard VLAs without sacrificing speed!

Training Strategies for Efficient Embodied Reasoning

Chen, Belkhale, Mirchandani, Mees, Driess, Pertsch, Levine. CoRL, 2025.

Embodied Reasoning so far has used a **fixed compute budget**

Can we do better **by letting it think longer at test time?**

→ **Test-Time Compute Scaling**

Isn't LLM Test-Time Compute Scaling just Planning?



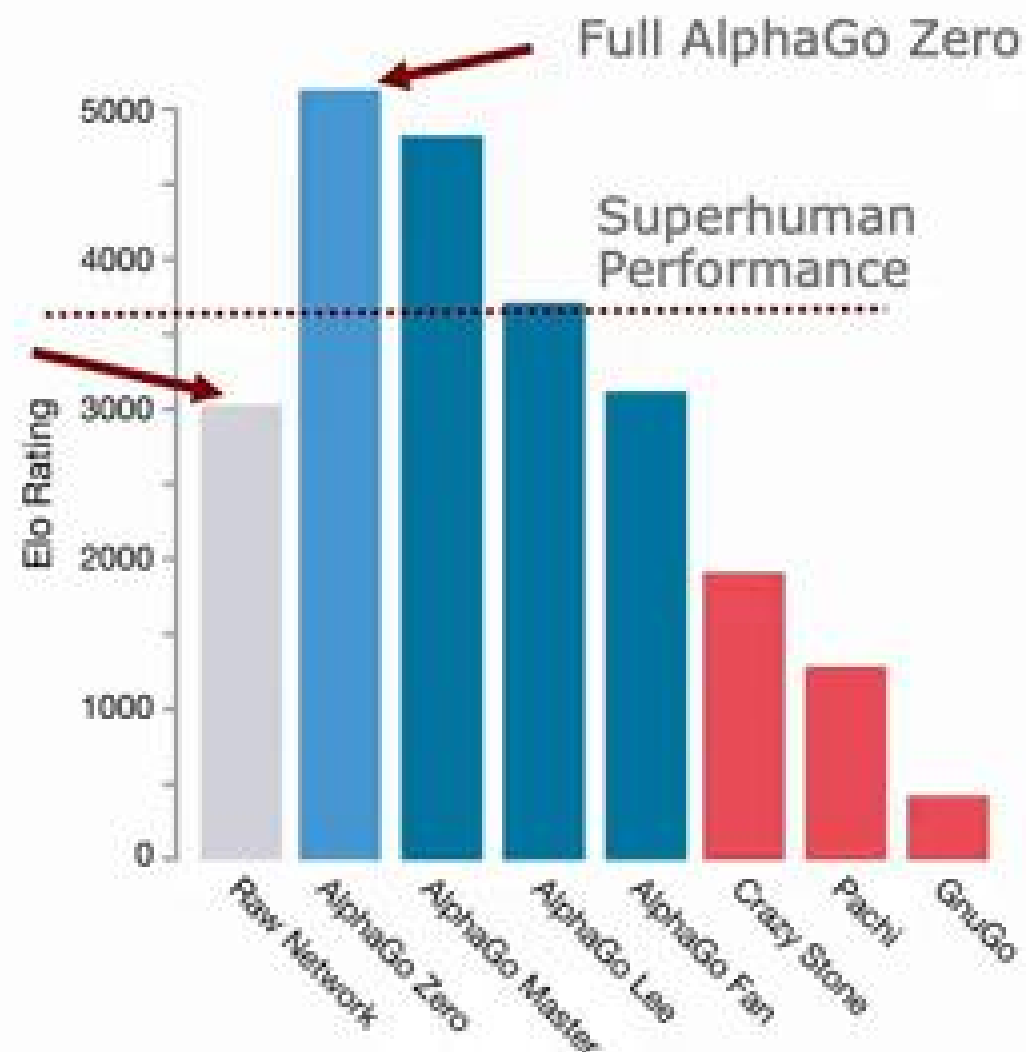
Scaling Test-Time Compute in Go

Rule of Thumb, increasing 120 Elo points requires either:

- 2x model size
- 2x test-time search

No test-time search

Improving the raw policy from 3000 to 5200 Elo points would require scaling the model by ~100,000x

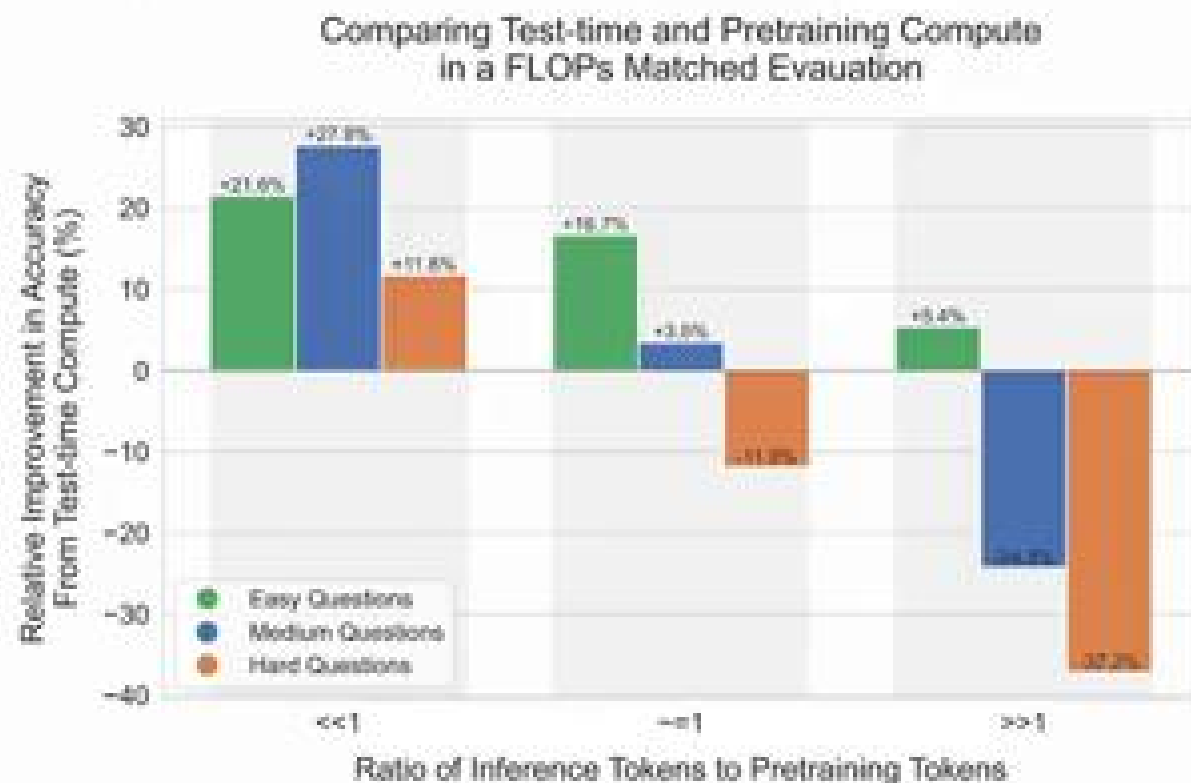


Mastering the Game of Go without Human Knowledge
Silver, et al. Nature, 2017.

Does this hold too in **language**?

Scaling Test-Time Compute in Language

- Compute-optimal test-time scaling beats a $\sim 14\times$ larger model on easy/medium problems at equal FLOPs



- SFT on own failed attempts + learned verifier — model learns to produce better answer given previous wrong ones, verifier picks the best one
- Generate multiple answers at test time, select best with verifier

Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters

Snell et al, 2024.

Why Do Reasoning Tokens Help?

- Any problem solvable in T computational steps can be solved by a **constant-size transformer** generating $O(T)$ reasoning tokens
- Generating $O(T)$ intermediate tokens before the answer gives the model $O(T)$ effective computation steps
- Without reasoning tokens: needs a much deeper/larger model or can't solve the problem at all

Chain-of-Thought with LLMs

More tokens = more computation steps

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

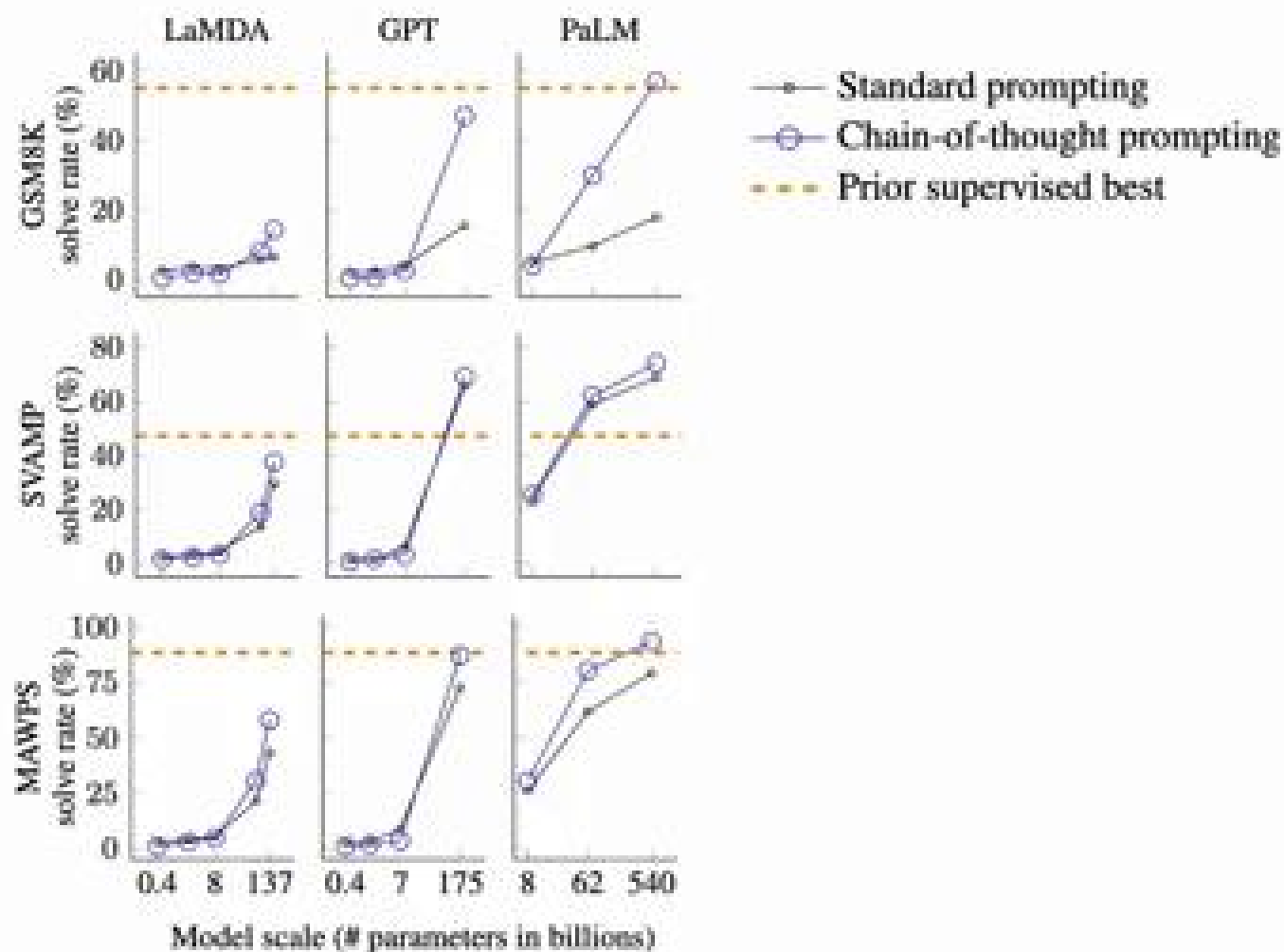
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models
Wei, Wang et al, NeurIPS 2023.

CoT Emerges in Large Models



Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Wei, Wang et al, NeurIPS 2023.

Original CoT Prompting = Few-shot/In-Context Learning

Q: [problem 1]

A: [step 1... step 2... therefore answer] ✓

Q: [problem 2]

A: [step 1... step 2... therefore answer] ✓

Q: [new problem]

A: ???

⚠ Prompt-sensitive, brittle

⚠ Requires Human input

⚠ Doesn't generalize beyond the examples

Instead of eliciting reasoning with human prompts, can we **train the model to reason intrinsically**?

Problem:

James writes a 3-page letter to 2 different friends twice a week. How many pages does he write a year?

Large Language Model (LLM) GPT-4o's Response:

Single Chain
of Thought

To solve this problem step-by-step, we can break it down as follows:

1. Determine Pages per Letter: James writes a 3-page letter.
2. Determine Number of Friends: He writes this letter to 2 different friends.
3. Calculate Pages per Week:
 - James writes to each friend twice a week.
 - Pages per letter: (3)
 - Letters per friend per week: (2)
 - Friends: (2)
 - Total pages per week = 3 pages/letter $\times$ 2 letters/week $\times$ 2 friends = 12 pages/week.
4. Calculate Pages per Year:
 - There are 52 weeks in a year.
5. Total pages per year = 12 pages/week $\times$ 52 weeks/year = 624 pages/year. Thus, James writes a total of 624 pages a year.

Large Reasoning Model (LRM) DeepSeek-R1's Response:

Thinking
Process

<think>

Okay, let me try to figure out this problem. So, James is writing a 3-page letter to two different friends twice a week. We need to find out how many pages he writes in a year. Here, let's break this down step by step. First, let me understand the details. [...]

Wait, if he writes to 2 friends twice a week, does that mean he writes 2 letters each time, and he does this twice? [...] So per week, he writes 2 friends $\times$ 3 pages $\times$ 2 times. Let me check that. [...]

Alternatively, maybe it's 3-page letter to 2 different friends each time he writes, and he does that twice a week. So each time he writes, he's writing two letters (each 3 pages), so that's $2 \times 3 = 6$ pages per writing session. Then, twice a week would be $6 \times 2 = 12$ pages per week. Then, multiply that by the number of weeks in a year. But how many weeks are in a year? Typically, 52 weeks. So $12 \text{ pages/week} \times 52 \text{ weeks} = 624$ pages. Is that right?

Wait, let me confirm. So, per week: [...] That makes sense.

But maybe I need to double-check if the problem is interpreted differently: [...] That seems consistent.

Wait, but wait a second. Is there any chance that the problem might be considering a year as 52 weeks, but some people might account for leap years or something else? But generally, in problems like this, unless specified, we take 52 weeks per year. So that's standard.

So, conclusion: [...] So the answer is 624. So I think that's correct.

</think>

James writes 3-page letters to 2 friends twice a week. Here's the breakdown:

[...]

Answer: James writes 624 pages in a year.

DeepSeek-R1 Thoughtology: Let's think about LLM reasoning
Marjanović et al, 2026.

Learning to Reason with RL - GRPO

- Sample answers, compare with each other, update toward good ones
- No reward model, no value function, no human labels



DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

Shao et al, 2024.

Why didn't this work before?

Why didn't this work before?

Base LLM needs to be good enough to produce any correct rollouts to reinforce!

Reasoning Behaviors Emerge from RL

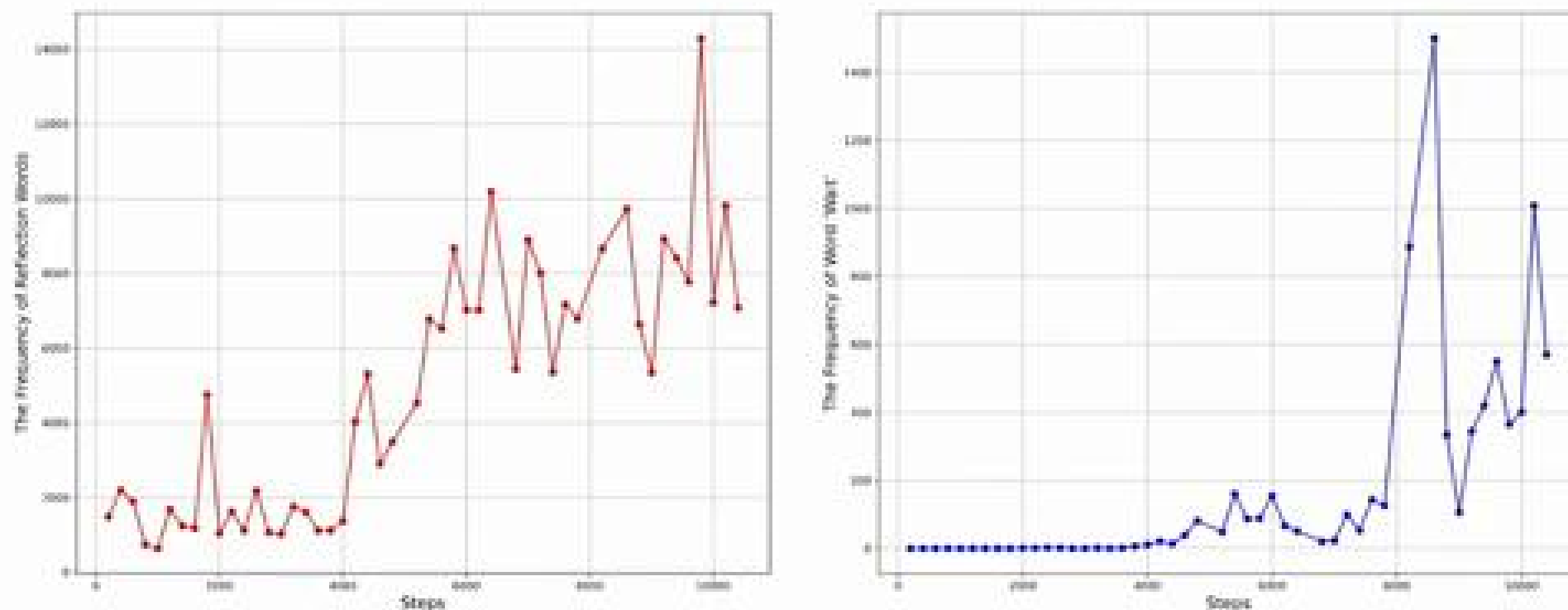


Figure 9 | Evolution of reasoning behaviors during training. (a) Frequency of representative reflective words during the training process; (b) Specific occurrence patterns of the word "wait" throughout the training process.

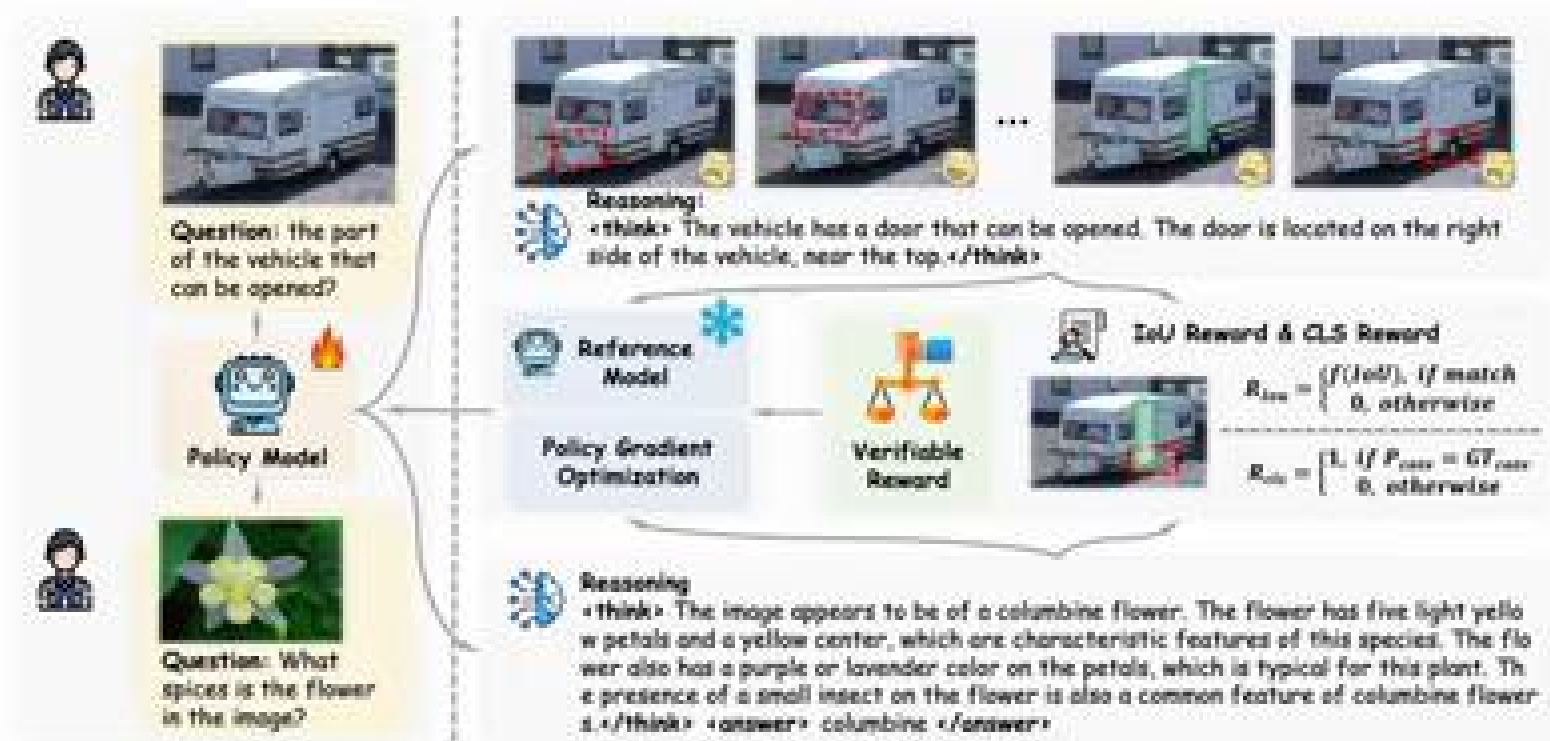
DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI, 2026.

Does this RL reasoning apply too in **vision**?

Visual Reinforcement Fine-Tuning

- Applies GRPO to enhance visual grounding of VLMs
- The verifier is IoU (geometric), not a math grader



Visual-RFT: Visual Reinforcement Fine-Tuning

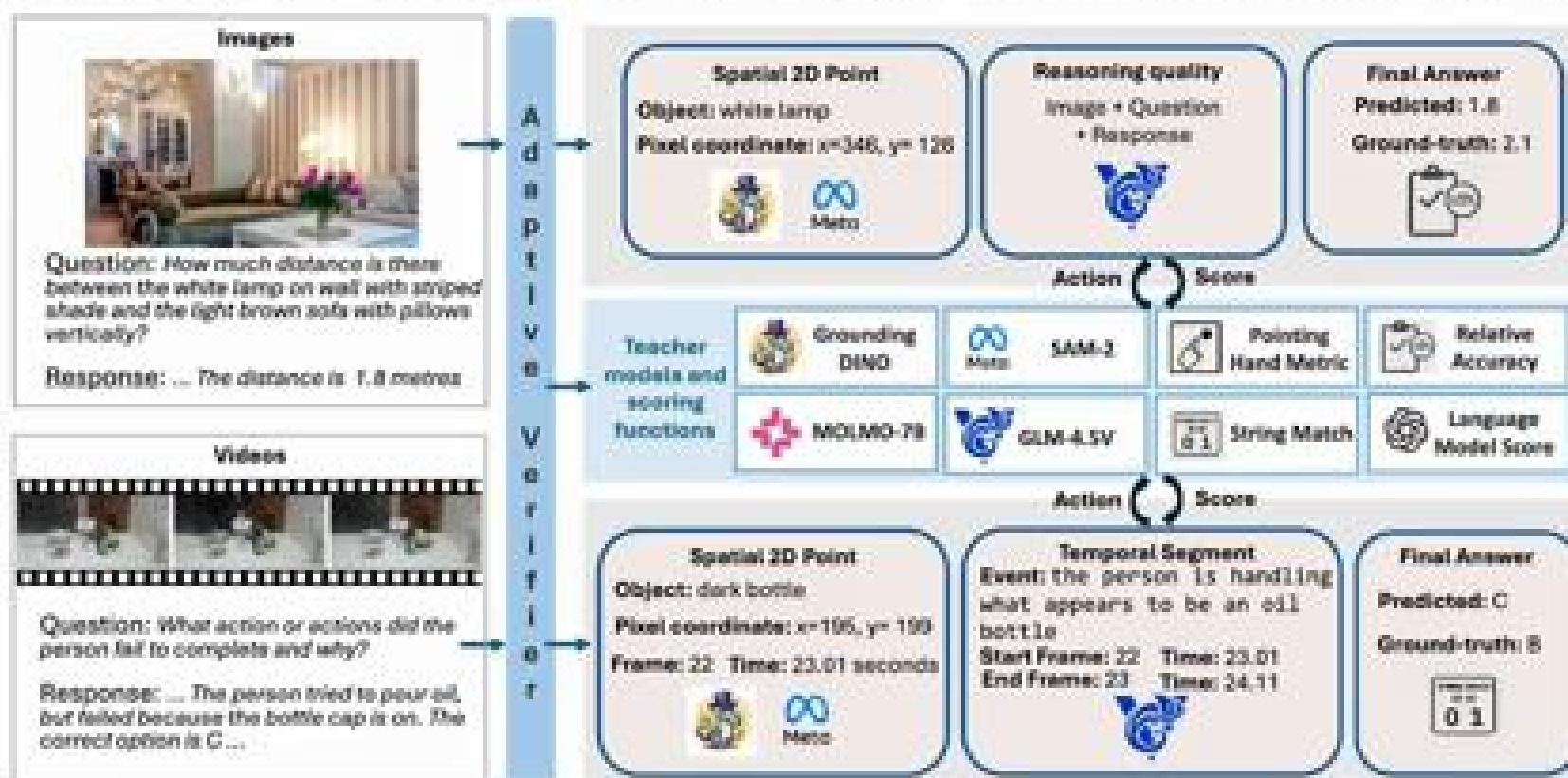
Liu et al., 2025.

Failure mode: Standard GRPO rewards the final answer only,
the **reasoning trace is invisible to the verifier**

In vision, hallucinated reasoning & correct answers aren't
mutually exclusive

ARGOS – Agentic Reasoning with Multimodal RL

- **Key:** aggregate multiple reward signal from intermediate reasoning steps, ie spatial & temporal reasoning in videos



Multimodal Reinforcement Learning with Agentic Verifier for AI Agents

Tan, Peng, Yang, Cheng, Mees et al. arxiv, 2025.

ARGOS - Embodied AI Benchmarks

- Test-time reasoning helps for high-level planning
- 3-4× better than supervised CoT on the hardest tasks
- For continuous control, internalize reasoning during VLA training and drop it at inference — ECoT-Lite style



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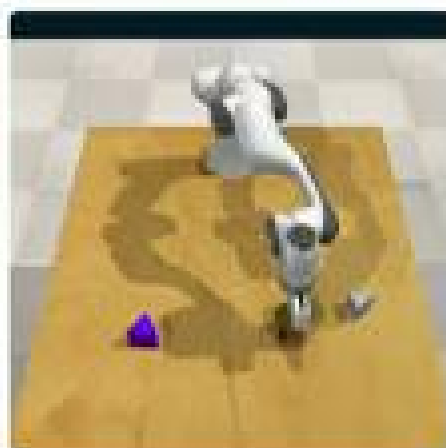


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Conclusion

- Scaling test-time compute is often more effective than scaling pre-training
- Reasoning SFT in VLAs improves representations for control — drop reasoning at inference
- RL can induce emergent reasoning in LLMs and VLMs
- For multimodal agents, verifying the reasoning trace, not just the final answer, is essential

Thank you for your attention